ML Model Deployment on EC2 and Access it

Task 1 - Display different classification models' accuracy_score, precision_score, recall_score, and f1_score and display it in a tabular form

Model Performance Evaluation:

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.95855	0.956009	0.95855	0.955330
1	Decision Tree	0.95505	0.954850	0.95505	0.954948
2	Random Forest	0.96820	0.966865	0.96820	0.966417
3	Gradient Boosting	0.97245	0.972937	0.97245	0.970133
4	SVM	0.95960	0.958052	0.95960	0.955490
5	KNN	0.95920	0.957349	0.95920	0.955232
6	Naive Bayes	0.90720	0.920187	0.90720	0.912697

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Task 2 - Create an EC2 instance

Description - *required* | Info

launch-wizard-3 created 2025-11-30T14:57:02.885Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) — Remove

Type	Protocol	Port range
ssh	TCP	22

Source type	Source	Description - *optional*
Anywhere	Add CIDR, prefix list or security group 0.0.0.0/0	e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 8000, 0.0.0.0/0, FastAPI / Uvicorn access) — Remove

Type	Protocol	Port range
Custom TCP	TCP	8000

Source type	Source	Description - *optional*
Anywhere	Add CIDR, prefix list or security group 0.0.0.0/0	FastAPI / Uvicorn access

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

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Task 3 - Connect to EC2 from your laptop using **Command Prompt**

scp -i mykey.pem app.py requirements.txt diabetes_model.pkl ubuntu@<EC2_IPv4>:/home/ubuntu/

```
C:\Personal\AUPP + FHSU\AUPP 2025\AUPP Fall 2025\CLD 375 (Cloud Machine Learning Foundation and Data Engineering)\A
ssignment\ML Model Deployment on EC2 and Access it>scp -i mykey2.pem app.py requirements.txt diabetes_model.pkl ubu
ntu@98.84.29.40:/home/ubuntu/
app.py                                                                    100% 1832     6.3KB/s   00:00
requirements.txt                                                          100%   57     0.2KB/s   00:00
diabetes_model.pkl                                                        100%  129KB 124.6KB/s   00:01
```

ssh -i "mykey.pem" ubuntu@<EC2_IPv4>

```
Last login: Sun Nov 30 14:32:31 2025 from 175.100.10.210
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-64-209:~$
```

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Task 4 - Install necessary updates/software to run your API

1). update + install python3 venv

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3-pip python3-venv git
```

2). create venv and install requirements

```
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

requirements.txt

necessary libraries:

- fastapi
- uvicorn[standard]
- boto3
- scikit-learn==1.6.1
- pydantic

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Task 5 - Convert your model to a pickle file

```
import pickle
from sklearn.ensemble import GradientBoostingClassifier

with open("diabetes_model.pkl", "wb") as f:
    pickle.dump(gb_model, f)
print("Saved diabetes_model.pkl")
```

```
Saved diabetes_model.pkl
```

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Task 6 - Create your API file

```python
# app.py
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
import pickle, boto3, json, uuid, os
from datetime import datetime, timezone

# ========== CONFIG ==========
MODEL_PATH = "diabetes_model.pkl"
S3_BUCKET = "orpf-prediction-bucket"
AWS_REGION = "us-east-1"
# ============================

class Person(BaseModel):
    # put feature names exactly as the model expects
    age: float
    bmi: float
    blood_glucose_level: float
    HbA1c_level: float
    hypertension: float
    heart_disease: float

app = FastAPI()

# Load model once on startup
with open(MODEL_PATH, "rb") as f:
    model = pickle.load(f)

# derive class info for human-readable output
classes = list(getattr(model, "classes_", []))
if set(classes) == {0, 1}:
    CLASS_MAP = {0: "non-diabetic", 1: "diabetic"}
else:
    CLASS_MAP = {c: str(c) for c in classes}

# Comment out the below line if not saving to S3
s3 = boto3.client("s3", region_name=AWS_REGION)

def upload_json_to_s3(obj: dict, key_prefix="logs/"):
    key = f"{key_prefix}{datetime.now(timezone.utc).isoformat()}_{uuid.uuid4().hex}.json"
    s3.put_object(Bucket=S3_BUCKET, Key=key, Body=json.dumps(obj).encode("utf-8"))
    return key

@app.post("/predict")
def predict(p: Person):
    try:
        # convert p to model input shape
        # Example: if model expects a 2D list
        X = [[p.age, p.bmi, p.blood_glucose_level, p.HbA1c_level, p.hypertension, p.heart_disease]]
        pred = model.predict(X)[0]
        prob = model.predict_proba(X)[0][1] if hasattr(model, "predict_proba") else None # Just class 1 probability
        prob_percentage = round(prob * 100, 2) if prob is not None else None
        label_name = CLASS_MAP.get(pred, str(pred))

        record = {
            "input": p.model_dump(),
            "prediction": int(pred),
            "label": label_name,
            "probability": prob,
            "timestamp": datetime.now(timezone.utc).isoformat()
        }

        upload_key = upload_json_to_s3(record)

        return {"result": int(pred), "label": label_name, "probability": f"{prob_percentage}%", "s3_key": upload_key}
        # return {"result": int(pred), "label": label_name, "probability": f"{prob_percentage}%"}
    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))
```

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Task 7 - Upload/deploy it to the EC2 instance

Task 8 - Run your api inside EC2 instance

Task 9 - Using Postman, send requests to the API

```
ubuntu@ip-172-31-72-1: ~

(venv) ubuntu@ip-172-31-72-1:~$ uvicorn app:app --host 0.0.0.0 --port 8000
INFO:     Started server process [13517]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     203.176.133.182:61458 - "GET /docs HTTP/1.1" 200 OK
INFO:     203.176.133.182:61458 - "GET /openapi.json HTTP/1.1" 200 OK
/home/ubuntu/venv/lib/python3.10/site-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feat
ure names, but GradientBoostingClassifier was fitted with feature names
  warnings.warn(
/home/ubuntu/venv/lib/python3.10/site-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feat
ure names, but GradientBoostingClassifier was fitted with feature names
  warnings.warn(
INFO:     203.176.133.182:61458 - "POST /predict HTTP/1.1" 200 OK
/home/ubuntu/venv/lib/python3.10/site-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feat
ure names, but GradientBoostingClassifier was fitted with feature names
  warnings.warn(
/home/ubuntu/venv/lib/python3.10/site-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feat
ure names, but GradientBoostingClassifier was fitted with feature names
  warnings.warn(
INFO:     203.176.133.182:52237 - "POST /predict HTTP/1.1" 200 OK
```

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Task 10 - Display prediction in the postman

Person 1

Input

```
{
  "age": 50,
  "bmi": 32.35,
  "blood_glucose_level": 80,
  "HbA1c_level": 6.23,
  "hypertension": 1,
  "heart_disease": 1
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://3.238.242.245:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "age": 50,
  "bmi": 32.35,
  "blood_glucose_level": 80,
  "HbA1c_level": 6.23,
  "hypertension": 1,
  "heart_disease": 1
}'
```

Request URL

```
http://3.238.242.245:8000/predict
```

Server response

Code | Details

200

Response body

```
{
  "result": 0,
  "label": "non-diabetic",
  "probability": "99.75%",
  "s3_key": "logs/2025-12-02T15:20:06.124245+00:00_2941382e5e7c433b8215212ed8ef92ac.json"
}
```

Output

Download

Response headers

```
content-length: 141
content-type: application/json
date: Tue,02 Dec 2025 15:20:05 GMT
server: uvicorn
```

Person 2

Input

```
{
  "age": 30,
  "bmi": 26.15,
  "blood_glucose_level": 90,
  "HbA1c_level": 4.23,
  "hypertension": 0,
  "heart_disease": 0
}
```

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://3.238.242.245:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "age": 30,
  "bmi": 26.15,
  "blood_glucose_level": 90,
  "HbA1c_level": 4.23,
  "hypertension": 0,
  "heart_disease": 0
}'
```

Request URL

```
http://3.238.242.245:8000/predict
```

Server response

Code | Details

200

Response body

```
{
  "result": 0,
  "label": "non-diabetic",
  "probability": "99.78%",
  "s3_key": "logs/2025-12-02T15:23:00.614397+00:00_02f7b48cb4df4cd6b8f985249172944f.json"
}
```

Output

Download

Response headers

```
content-length: 141
content-type: application/json
date: Tue,02 Dec 2025 15:23:00 GMT
server: uvicorn
```

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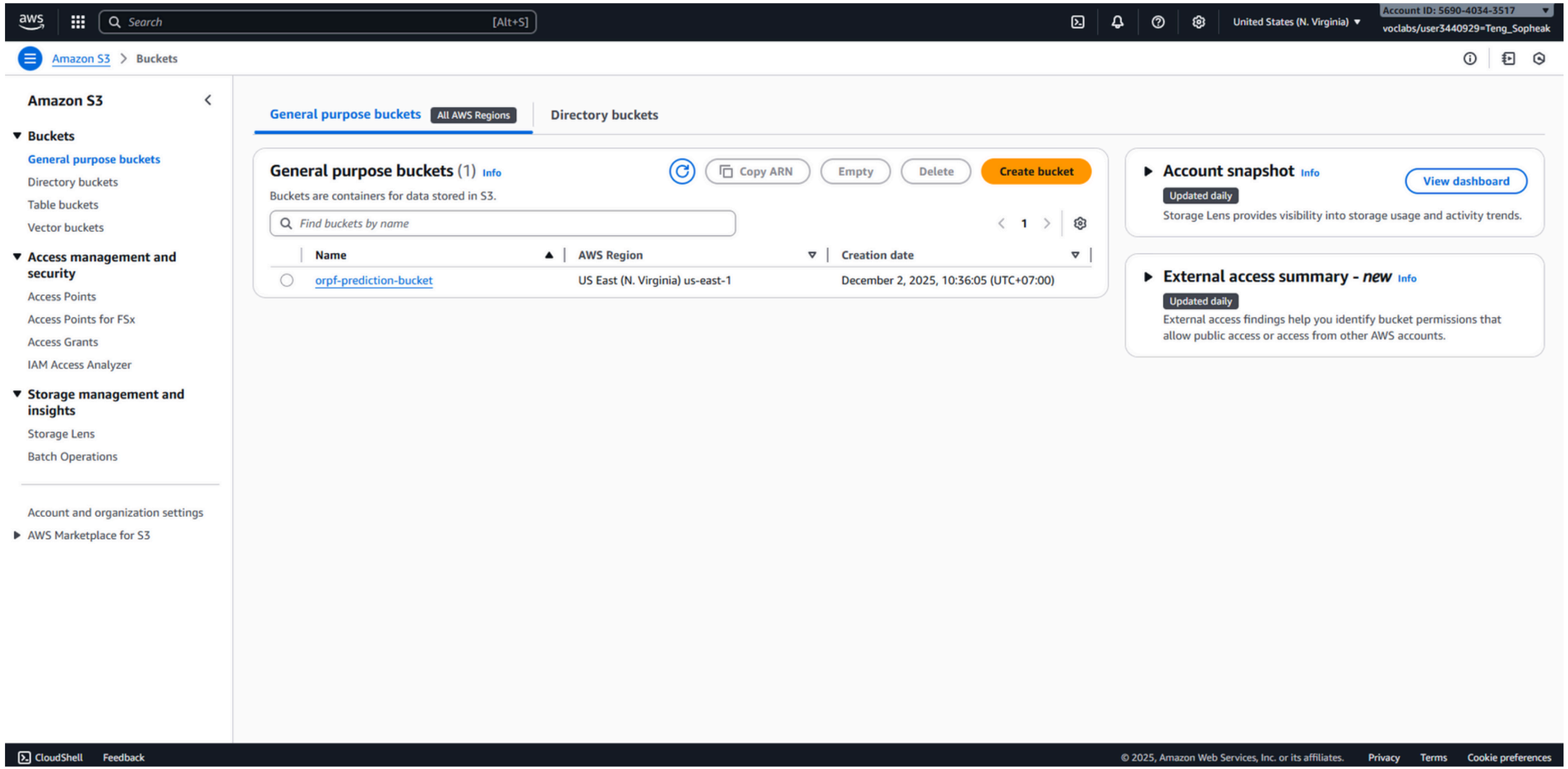
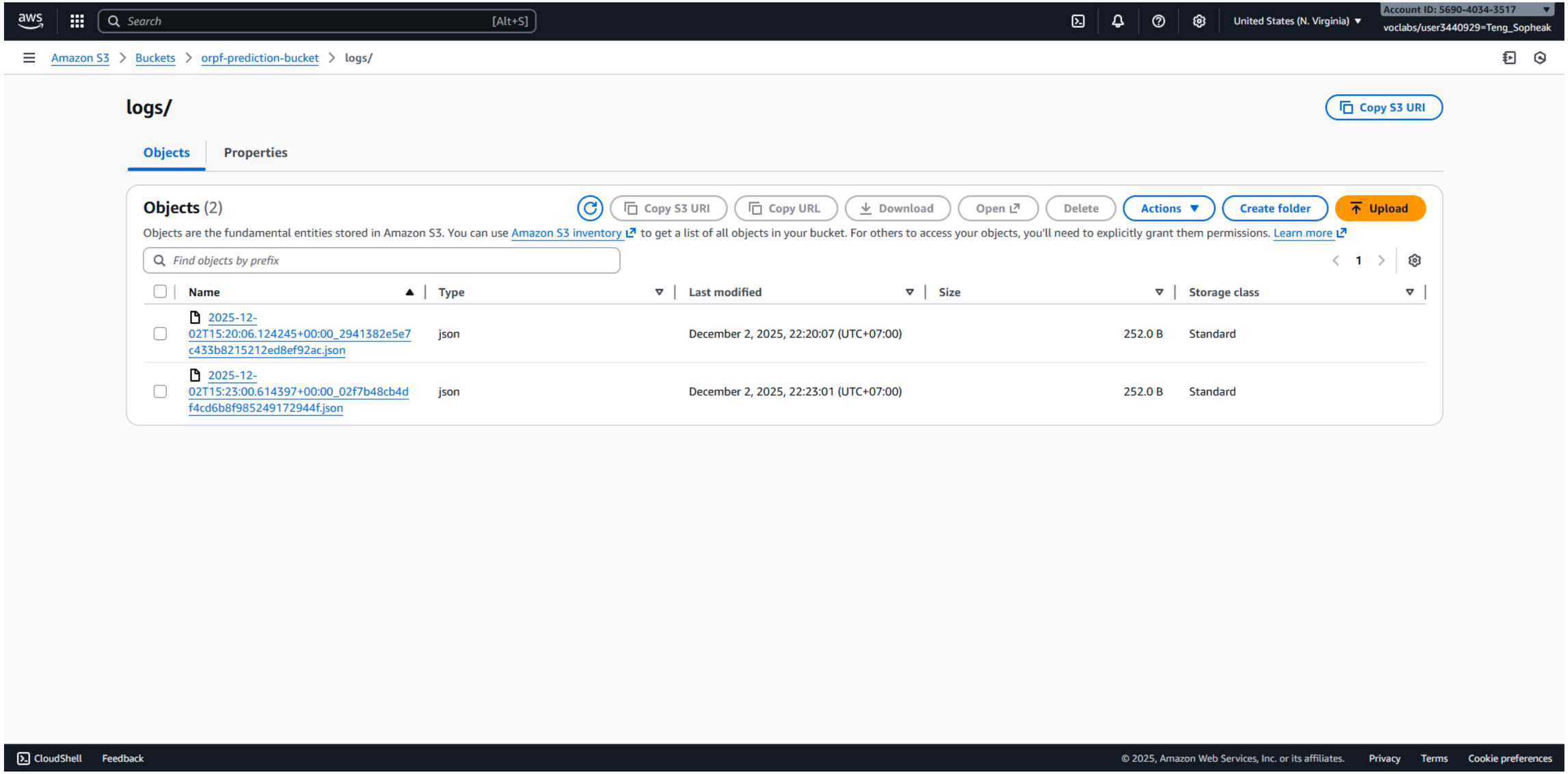
Task 11 - Create an S3 bucket



ML Model Deployment on EC2 and Access it

Task 12 - Try to store the input data and output prediction in the S3 bucket